

Shubhangi Agarwal

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SUMMARY

I am a researcher with an interest in **Graphs**. I have developed various algorithms based on statistical analysis for **Subgraph Querying** in large complex graphs. Some of the many areas that can benefit from the querying of subgraph structures are information extraction, recommendation systems, disease diagnostics, fraud detection. During my post-doc, I worked on **HyGraphs**, where as a team we explored development of a novel data model combining property graphs and time series data models, with an aim to provide a single platform for seamless, intuitive and efficient processing of large scale datasets which are dynamic in nature.

EDUCATION

Ph.D. in Computer Science and Engineering

Indian Institute of Technology Kanpur, Uttar Pradesh, India

CGPA: 8.25

2014 - 2023

Bachelor of Technology Computer Engineering

Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat, India

CGPA: 8.71

2010 - 2014

PHD THESIS

Subgraph Matching and Mining in Large Graphs

Supervisor: Arnab Bhattacharya

- Developed algorithms for Approximate Subgraph Matching in both deterministic and probabilistic graphs.
- Proposed a Graph Neural Network model for robust node embeddings with positional information.

PUBLICATIONS

- “A Case Study of Cross-Lingual Zero-Shot Generalization for Classical Languages in LLMs”, V S D S Mahesh Akavarapu, Hrishikesh Terdalkar, Pramit Bhattacharyya, **Shubhangi Agarwal**, Vishakha Deulgaonkar, Chaitali Dangarikar, Pralay Manna, and Arnab Bhattacharya, 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025, Austria.
- “Towards Hybrid Graphs: Unifying Property Graphs and Time Series”, Mouna Ammar, Christopher Rost, Riccardo Tommasini, **Shubhangi Agarwal**, Angela Bonifati, Petra Selmer, Evgeny Kharlamov and Erhard Rahm, 28th International Conference on Extending Database Technology (EDBT), 2025, Spain.
- “Transforming Time Series into Graphs and Back with HyGraph”, Mouna Ammar, **Shubhangi Agarwal**, Angela Bonifati and Erhard Rahm, Proceedings of the Workshops of the EDBT/ICDT Joint Conference, 2025, Spain.
- “VeNoM: Approximate Subgraph Matching with Enhanced Neighbourhood Structural Information”, **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, 7th Joint International Conference on Data Science and Management of Data (CODS-COMAD), 2024, India.
- “VerSaChI: Finding Statistically Significant Subgraph Matches using Chebyshev’s Inequality”, **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, Proceedings of the International Conference on Information and Knowledge Management (CIKM), 2021, pages 2812-2816, Australia.
- “GraphReach: Position-Aware Graph Neural Network using Reachability Estimations”, Sunil Nishad, **Shubhangi Agarwal**, Arnab Bhattacharya and Sayan Ranu, Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI), 2021, pages 1527-1533, Canada.
- “ChiSeL: Graph Similarity Search using Chi-Squared Statistics in Large Probabilistic Graphs”, **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, Proceedings of the International Conference on Very Large Data Bases (VLDB), 2020, pages 1654-1668, Japan.

EXPERIENCES

Project Executive Officer

BharatGen, IIT Kanpur, India

Sep 2025 - Present

- Project: ‘BharatGPT: A Suite of Generative AI Tech for India’.
- Funding Agency: Department of Science and Technology (DST), Government of India

Post-doctoral Researcher

LIRIS Research Lab, CNRS, Université Claude Bernard Lyon 1, France

May 2024 - Apr 2025

- Project: 'HyGraph: Querying and Analytics on Hybrid Graphs'.
- Funding Agency: French Agence Nationale de la Recherche (ANR) and the Deutsche Forschungsgemeinschaft (DFG; German Research Foundation)

Senior Student Research Associate

Aug 2022 - Dec 2023

IIT Kanpur, India

- Project: 'Short and Long-term Fog Prediction for North Indian Cities'.
- Funding Agency: Scheme for Transformational and Advanced Research in Science (STAR), Ministry of Education, Government of India.

Summer Intern

Jun 2015 - Jul 2015

Flipkart Internet Private Limited

- Worked in a project to prepare a model for identification of 'fake' listings.

TEACHING

Instructor

Sep 2024 - Dec 2024

Université Claude Bernard Lyon 1, France

- Program: M2 - Data and Intelligence for Smart Systems (DISS)
- Course: Data Processing and Analytics (Jointly with others)

Teaching Assistant

Aug 2014 - Apr 2021

IIT Kanpur, India

- Graded and evaluated projects for various courses of Computer Science.

Senior Tutor

Aug 2017 - Apr 2018, Aug 2019 - Apr 2020

IIT Kanpur, India

- Led teams of strength ~60; Assisted in backend management, paper-setting and grading.

Teaching Assistant for MOOC

July 2017 - Sep 2017

NPTEL - Remote

- Crafted objective questions and resolved student queries on Fundamentals of Database Systems.

ADMINISTRATIVE EXPERIENCES

Program Committee Member

2025 - current

- GRADES-NDA Workshop - SIGMOD/ PODS (2025), ICWE (2025), TGD Workshop - EDBT/ ICDDT (2025).

External Reviewer

2020 - current

- WSDM (2024), CIKM (2021, 2022), DASFAA (2022), CoDS-COMAD (2020, 2021), KDD (2021).

TECHNICAL SKILLS

Languages : C, C++, Python, Java, R, JavaScript, PHP, Shell scripting
Databases : MySQL, Neo4j
Libraries : PyTorch, Tensorflow, Scikit-learn, Numpy, Pandas
Tools : git, LaTeX, Docker, Weka

REFERENCES

Arnab Bhattacharya

Professor, Indian Institute of Technology Kanpur, India

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Sourav Dutta

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Angela Bonifati

Distinguished Professor, Lyon 1 University, France

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Riccardo Tommasini

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